Functional form or Machine-Learning-Based Ground-Motion Models? An application to the Italian dataset

Lucia Luzi (lucia.luzi@ingv.it)
Istituto Nazionale di Geofisica e Vulcanologia https://orcid.org/0000-0003-4312-580X
Chiara Felicetta
Istituto Nazionale di Geofisica e Vulcanologia
Giovanni Lanzano
Istituto Nazionale di Geofisica e Vulcanologia

Research Article

Keywords:

Posted Date: June 14th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-3025369/v1

Abstract

This paper examines the advantages and drawbacks of the use of a functional form in empirical ground-motion modelling compared to machine learning algorithms.

Typically, models based on linear regression and predefined functional forms have limits in representing complex nonlinear behaviour of source, attenuation and site effects present in the data. We investigate the efficiency of different machine learning algorithms using the dataset of Italian strong motion records, consisting of 5,607 records relative to 146 earthquakes and 1,657 stations, employed to derive the most recent ground motion model for Italy. We quantify the differences in the predictive capabilities of both approaches in terms of standard deviation, which is broken down into between-event, between-station and event- and site-corrected components, implemented as random effects.

When datasets are sufficiently large, various ML algorithms tend to provide more accurate estimates compared to the conventional linear regression-based method and the Gaussian Process Regression is the best-performing algorithm with the Italian dataset. The conventional regression-based methods are a better tool when limited data is available.

Introduction

This paper aims a comparison between traditional methods for predicting ground motion Intensity Measures (IMs), usually based on linear regression, and machine learning algorithms (algorithms to analyze huge datasets), which are widely used today for different purposes thanks to the increasing availability of data and computing performance.

Generally speaking, Ground Motion Models (GMMs) are empirical equations to estimate IMs as a function of several variables, generally related to the seismic source (e.g. magnitude), source-to-site distance (e.g. epicentral, hypocentral, Joyner-Boore distance, among others) and local site conditions (e.g. site categories or averaged shear wave velocities in the uppermost 30 meters), as described thoroughly by Douglas and Edwards (2016). Simple functional forms and/or few variables are not always capable to model all aspects of ground motion, therefore several additional variables can be added to the linear regression models to reduce the variability associated with the prediction, although many of them are usually difficult to obtain (e.g. style of faulting, depth to the top of rupture, sediment thickness, hanging wall location, see Chiou and Youngs, 2014; Campbell and Bozorgnia, 2014).

An alternative to regression in ground motion prediction could be the use of machine learning (ML) algorithms, non-parametric models that are fully trained by data that have recently been employed successfully by many authors. In general, the learning algorithm builds a model from a training set of observations and then evaluates the predictive performance of the model on an independent dataset that should represent new data. The use of ML approaches dates back to the early 2000s when Kerh and Ting (2005) selected a combination of seismic parameters including epicentral distance, focal depth, and magnitude from historical records at 30 stations to apply the back-propagation neural network model, to estimate peak ground acceleration at ten train stations along the high-speed rail system in Taiwan. Güllü and Erçelebi (2007) applied the neural network approach for the prediction of Peak Ground Acceleration (PGA) using about 1000 strong motion data from Turkey. Derras et al (2012) applied the artificial neural network approach to a large subset of the KiK-net seismic database, which includes 3,891 records from 398 sites and 335 earthquakes to predict horizontal PGA. More recently, machine learning algorithms have been applied to other ground motion intensity measures, such as PGV (Peak Ground Velocity), and the 5% damped spectral response acceleration values (SA) at periods of interest for structural design. Derakhshani and Foruzan (2019) developed a model for peak ground acceleration (PGA), PGV, and PGD based on a deep neural network (DNN), using the Next Generation Attenuation dataset for the western United States (NGA-West2), which includes 21,336 ground-motion records, demonstrating the outperformance for traditional regression-based or machine learning models by the improvement of correlation coefficients between observed and predicted intensity measures. Khosravikia and Clayton (2021) applied Artificial Neural Network, Random Forest and Support Vector Machine to a dataset of 4,528 ground motions of earthquakes in Oklahoma, Kansas, and Texas; Seo et al (2022) used three machine-learning techniques (e.g., artificial neural network, random forest, and gradient boosting) to 1,189 ground motions from 77 earthquakes in Korea; Mori et al (2022) applied ML techniques to produce shaking scenarios, in alternative to USGS Shakemap, using 15,779 records from the Engineering Strong Motion and the Italian Strong Motion databases (Luzi et al, 2016; Luzi et al, 2019).

Seismological datasets are constantly increasing in terms of available waveforms, due to the fast development of new networks and strong motion databases, therefore recent studies, such as Kim et al (2020) or Okazaki et al (2021) developed models using much larger datasets than previous studies. Kim et al (2020) used 115,501 records of the Kiban-Kyoshin network (KiK-Net) database to predict ground motion amplifications, while Okazaki et al (2021) used 234,552 records of the same KiK-Net database, to derive peak parameters. In general, the use of ML algorithms results in a more accurate estimation of ground motion.

The estimation of intensity measures using ML is performed by selecting the same predictor variables as in regression-based models, that reproduce the contributions of source (e.g. magnitude, focal mechanism), attenuation (e.g. epicentral, hypocentral, Joyner-Boore or rupture distance) and site ($V_{S,30}$ or soil categories). The attention has been concentrated on the improvement of the prediction performance based on residuals between observed and predicted IMs (e.g. Root-Mean-Square Error, RMSE, which is the standard deviation of the residuals; Mean Absolute Error, which is the sum of absolute residuals divided by the sample size; Regression Coefficient, which describes the relationship between a predictor variable and the response). Little or no attention has been paid to the individuation of additional explanatory variables, that could have no direct connection with the physics of earthquakes.

This study has the ambition of investigating the potential of ML algorithms exploiting the dataset used by Lanzano et al (2019) to derive the most recent ground motion model for Italy and explore the different results, if any, to a classical regression model. Different ML algorithms are tested and their efficiency is assessed through the evaluation of the standard error. The results of the best model are compared with the empirical equation by Lanzano et al (2019), to explore and explain the differences. As in Lanzano et al (2019), the residuals between predictions and observations are used to identify the contribution to the variability of events (between-event), stations (between-stations) and event- and site-corrected residuals.

The framework and all the ML algorithms presented in this study are implemented using Matlab©.

Ground motion database and linear regression model

The training dataset is the flat file used by Lanzano et al (2019) to derive the most recent ground motion model for Italy which includes 5,607 records, relative to 146 earthquakes and 1,657 stations. The flat-file (Lanzano et al 2022), derived from the Engineering Strong-Motion (ESM) database (Luzi et al, 2020; Lanzano et al 2018), has been improved with the records of some regional small-magnitude events (Mw < 4:0), available in the ITalian ACcelerometric Archive (ITACA) database (Pacor et al, 2011; Russo et al, 2022) and, to increase the maximum usable magnitude, the records of 12 worldwide events in the magnitude range 6.1–8.0 (3 Turkey, 2 Japan, 2 New Zealand, 2 California, 1 Iceland, 1 Iran, and 1 Greece) have been added. The complete list of events is provided in Table S1 of the electronic Appendix of Lanzano et al (2019).

The dataset samples only active crustal regions in Italy and the maximum hypocentral depth of earthquakes is 30 km. Particular care has been spent on the earthquake selection to avoid record oversampling in Central Italy, an area struck by the Mw 6.0 Umbria-Marche 1997, Mw 6.2 L'Aquila 2009 and Mw 6.5 Central Italy 2016 seismic sequences.

The epistemic uncertainty that could affect the datasets of the past studies on the machine learning algorithms, that are assembled from different data sources, is reduced since i) the data are processed with the same technique (Paolucci et al 2011), ii) the intensity measures are calculated with the same algorithms and iii) the metadata of events and recording stations have been carefully revised.

A detailed description of the dataset and parameters used in this study is in Lanzano et al (2019).

Figure 1 sketches the distribution of the metadata of seismic records, in terms of earthquake magnitude, source-to-site distance, event depth and style of faulting and V_{S30} of recording sites (averaged shear wave velocity in the uppermost 30 m). From Fig. 1 the bulk of the data is recorded within 100 km from the source, the prevailing style of faulting is normal and earthquakes have hypocentral depths between 8 and 12 km. The subsoil of recording sites has a V_{S30} in the range 200–600 m/s and a very few sites are on rock. The largest data density is found in the magnitude range 4-4.5 and distance range 25–50 km (Fig. 2).

Lanzano et al (2019) calibrated a predictive model where the logarithm base 10 of the *IM* is the sum of the following terms:

Source term

$$F_M\left(M_W\right)=\{b_1(M_w-M_H)whereM_w\le M_h\}\ [1]$$

$$F_M\left(M_W\right)=\{b_2(M_w-M_H)whereM_w>M_h\}$$

$$F_M\left(SOF\right)=f_jSOF_j\ [2]$$

Eq. [1] models the ground motion with a linear dependence on moment magnitude and two different coefficients are hypothesized, depending on a period-dependent hinge magnitude M_h; the style of faulting (SOF) is treated as a dummy variable (f_j takes the value of 1 or 0 depending on the fault type: strike slip SS, thrust fault TF, or Normal Fault NF), and the regression is performed constraining to zero the coefficient for normal faulting.

Attenuation term

The attenuation of ground motion with distance R (in km) is the combination of the geometric attenuation (magnitude dependent) and anelastic attenuation; two distance metrics are considered, the rupture distance (km) and the Joyner-Boore distance (km).

$$F_D\left(M_{W,R}\right)=\left\{\left[c_1\left(M_w-M_{ref}\right)+c_2\right]log10\left(R\right)+c_3R\right\}\ [3]$$

$$R=\sqrt{\left({R_i}^2+h^2\right)}\ [4]$$

where subscript *i* can be either *RUP*, to indicate rupture distance, or *JB* to indicate Joyner-Boore distance; h is the pseudo-depth (km) resulting from the regression.

Site term

The site effects are accounted for according to a function of the averaged velocity in the uppermost 30m

$$F_S\left(V_{s30}\right)=klog10\left(V_0/800\right)\ [5]$$

where V_0 = V_{S30} when V_{S30} ≤ 1500 m/s and V_0 = 1500 m/s otherwise

Machine learning workflow

To derive a predictive model we used the Regression Learner tool (RLt), available in Matlab ©, which allows us to automatically train a selection of models or compare and tune options of linear regression models, regression trees, support vector machines, Gaussian Process Regression models and ensemble of trees. The different learning structures provide the opportunity of exploring different types of learning algorithms in extracting features that exist in the available data. The ground motion dataset has been split and used to train and validate different models. The most suitable model has been then selected based on the Root-Mean-Square Error (RMSE) and the distribution of the residuals (difference between the logarithm base 10 of observations and predictions); in addition, the trend of residuals in function of different predictors has been checked to verify the absence of biases.

We initially calculate the correlation matrix among M_W, $\log_{10}(R_{JB})$, SOF, $\log_{10}(V_{S30}/800)$ and H with several intensity measures (PGA, SA at T = 0.1s, SA at T = 1s and SA at T = 10s). Figure 3 shows the correlation matrix between couples of predicting variables. The considered IMs are mostly correlated with magnitude (positive correlation) and distance (negative correlation), while a weaker inverse correlation is found for hypocentral depth (H) and very small correlation with the style of faulting. As expected, a negative, significant, correlation is found with the soil amplification, whose proxy is the V_{S30}.

We select the same explanatory variables as Lanzano et al (2019) to train the model: moment magnitude (M_W), rupture or Joyner-Boore distance (R_{RUP} or R_{JB}), style of faulting (SOF) and averaged velocity in the uppermost 30 m (V_{S30}). In addition, we considered the event hypocentral depth (H), which was not considered as a separate variable by Lanzano et al (2019), since they obtained an averaged hypocentral depth (pseudo-depth) through regression.

We train a model for the same intensity measures (IMs) as in Lanzano et al (2019): peak ground acceleration (PGA), peak ground velocity (PGV), and 36 ordinates of the 5% damped acceleration response spectra (SA) in the period range 0.04s – 10 s. In particular, we consider the geometrical mean of the two horizontal components.

The dataset is randomly split into two parts, 70% as training data and 30% as test data, such that each class is correctly represented in the resulting subsets (the training and the test set). The selected percentages are the most commonly used for small datasets since they ensure a trade-off between the amount of information helpful to the training and the accurate error estimation on the test dataset. For the model evaluation, we use the k-fold cross-validation implemented in the Matlab-RLt. This method allows us to select the number of folds to partition the data into k disjoint validation sets. For each validation fold the tool trains a model using the training-fold observations and assesses the model performance using validation-fold data. The average validation error is calculated over all folds.

As the first step, we try to identify the most suitable model to predict the base 10 logarithm of peak ground acceleration (PGA), which is characterized by the largest variability in traditional regression models.

We explore the following predictive performance of different ML algorithms:

- Fine Tree: a fine regression tree (minimum leaf size is 4);
- Linear regression: a linear regression model with only intercept and linear terms;
- Linear SVM: a support vector machine that follows a simple linear structure in the data using the linear kernel;
- Quadratic SVM: a support vector machine that uses a quadratic kernel;
- Exponential GPR: a Gaussian exponential model that used exponential kernel;
- Squared Exponential GPR: a Gaussian process model that uses the squared exponential kernel.

Table 1 displays the performance of different models for Peak Ground Acceleration in terms of RMSE since the correlation coefficients between observations and predictions is larger than 0.8 for all algorithms. We find that the model best fitting data is the Gaussian Process Regression (GPR) with the Exponential Kernel (Table 1), in agreement with Mori et al (2022).

A Gaussian Process (GP) is a generalization of the Gaussian probability distribution; whereas a probability distribution describes random variables which are scalars or vectors (for multivariate distributions), a stochastic process governs the properties of functions (Rasmussen and Williams, 2006). GPR aims to move from the training dataset to a mean function *f* that makes predictions for all possible input values. The GPR assigns a prior probability to every possible function, where higher probabilities are assigned to functions that we consider to be more likely; when many data points are considered during the training process one would see that the mean function *f* adjusts itself to pass through the points so that the posterior uncertainty would reduce close to the observations.

The properties of functions are induced by the covariance function of the Gaussian process and the problem of learning is exactly the problem of finding suitable properties for the covariance function, that specifies the covariance between the two latent variables $f(x_i)$ and $f(x_j)$, where both x_i and x_j are *d-by-1* vectors and *d* is the number of observations. In other words, it determines how the response at one point x_i is affected by responses at other points

x_j, $i \neq j$, $i = 1, 2, ..., n$.

The covariance function $k(x_i,x_j)$ can be parameterized in terms of the kernel parameters in vector θ. Hence, it is possible to express the covariance function as $k(x_i,x_j \mid \theta)$.

For many standard kernel functions, the kernel parameters are based on the signal standard deviation σ_f and the characteristic length scale σ_l. The characteristic length scales briefly define how far apart the input values x_i can be for the response values to become uncorrelated. Both σ_l and σ_f need to be greater than 0, and this can be enforced by the unconstrained parametrization vector θ, such that

$\theta_1 = \log \sigma_l$, $\theta_2 = \log \sigma_f$ [6]

In our case, the most suitable kernel is the Exponential Kernel.

$$k(x_i, x_j|\theta) = \sigma^2 e^{\left(-\frac{r}{\sigma_l}\right)} \quad [7]$$

Where *l* is the characteristic length scale and

$$r = \sqrt{\left(x_i - x_j\right)^T \left(x_i - x_j\right)} \quad [8]$$

is the Euclidean distance between x_i *and* x_j

Table 1
Root-Mean-Square Error (RMSE) in function of ML algorithms for Peak Ground Acceleration.

Variables	
M_W, $\log_{10}(R_{JB})$, SOF, $\log_{10}(V_{S30}/800)$, H	
Algorithm	**RMSE**
Fine Tree	0.365
Linear SVM	0.388
Linear regression	0.388
Quadratic SVM	0.344
Exponential GPR	*0.307*
Squared Exponential GPR	0.328

In the second step, we calculate the RMSE as a function of the explanatory variables (Table 2) using the GPR algorithm, to understand the effect of increasing information by adding variables according to the degree of correlation with the selected IM, as shown in Fig. 3. It is evident how the RMSE progressively reduces when information is added to the training set: we initially have an RMSE equal to 0.33949 if we consider only magnitude and distance, and an RMSE of 0.30646 after introducing the event hypocentral depth, the style of faulting and the site V_{S30}.

Table 2
Root-Mean-Square Error (RMSE) in function of the variable combination for Peak Ground Acceleration.

Model	
Exponential GPR	
Variables	**RMSE**
M_W, $\log_{10}(R_{JB})$	0.33949
M_W, $\log_{10}(R_{JB})$, H	0.31961
M_W, $\log_{10}(R_{JB})$, H, $\log_{10}(V_{S30})$	0.31144
M_W, $\log_{10}(R_{JB})$, H, SOF, $\log_{10}(V_{S30})$	0.3076

If we consider the rupture distance as a metric and we account for magnitude, distance, depth, style of faulting and V_{S30} we obtain an RMSE equal to 0.30719, which is comparable to the result obtained considering the Joyner-Boore distance as the predictor variable.

To check the accuracy of the prediction, we calculated the residuals between the base 10 logarithms of observed and predicted IMs. Residuals are normally distributed and have a mean very close to zero, as shown in Fig. 4 for PGA and spectral acceleration at T = 1.0s.

The residuals are then partitioned into between-event, site-to-site and event- and site- corrected residuals (Al Atik et al, 2010), applying the random effects to stations and events, as in Lanzano et al (2019). The between-events residual corresponds to the average misfit of recordings from one particular earthquake to the median ground-motion model; the site-to-site residual (or site term) represents the average misfit at each station. Finally, the event- and site- corrected residual represents the variability that cannot be explained by the model (e.g. directivity effects, azimuth-dependent attenuation of seismic waves, radiation pattern).

The between- event, site-to-site and event- and site- corrected terms obtained in this study are normally distributed and their standard deviation is reported in Appendix A for all IMs. To verify the presence of any bias in the residuals we plot the between-event term in function of the earthquake magnitude, the site-to-site term in function of the V_{S30} and the event- and site-corrected term in function of the Joyner-Boore distance. Figure 5 displays the absence of any trend in the residuals, which indicates the robustness of the analysis.

In analogy with the ground motion model obtained by Lanzano et al (2019), which is termed ITA18, hereinafter we designate ITA18-ML the GMM obtained with the GPR algorithm.

We finally compare the between-event, the site-to-site and the total standard deviations of ITA18 and ITA18-ML, for each spectral period (Fig. 6). In general, the reduction of each of term is about 30%, with a significant decrease in the period interval 0.15s-0.8s.

Figure 7 displays the difference between the median values of ITA18 and ITA18-ML models for two selected magnitudes, Mw = 4.0 and Mw = 6.8. For this comparison the event depth in ITA18-ML is 10 km, that is close to the event pseudo-depth of ITA18 calibrated in function of the Joyner-Boore distance. The two models are also compared with observations, although a few of them are available for Mw 6.8 +/- 0.3 events. The mean model obtained with the ML algorithm adapts well to observations, therefore ITA18 and ITA18-ML overlap when a large number of observations are available for the combination magnitude – style of faulting - V_{S30} (in Fig. 7 magnitude 4.0, normal faulting and V_{S30} = 600 m/s). Surprisingly, the two approaches result in very similar predictions even when observations are scarce or absent (e.g. Mw = 6.8, thrust fault and V_{S30} = 600 m/s in Fig. 7).

In Fig. 8 the predictions of ITA18 and ITA18-ML for the same IM (SA at T = 1s), normal fault, V_{S30} = 600 m/s and two magnitudes (Mw = 6.8 or Mw = 4) are compared for two different event depths: 5km and 20 km. For this comparison, the distance metric is the rupture from the fault plane.

For shallow events (e.g. H = 5 km), the mean prediction of ITA18-ML and ITA18 for the two selected magnitudes are comparable, that is the mean prediction of ITA18-ML lies between the mean plus/minus one standard deviation of ITA18 (although the Mw 6.8-5 km prediction could be meaningless because of the absence of observations, we display the result for completeness). In the case of deep events (e.g. H = 20 km), the predicted ground motion in the near-fault by ITA18-ML is substantially lower than ITA18 at both magnitudes. We would like to evidence this result since ground motion models based on classic regression could overpredict ground motion of deep events in the near-source, even if they account for the hypocentral depth through the rupture distance, whereas models calibrated on the Joyner – Boore distance cannot capture at all the effect we have described, overestimating the ground motion of deep events near the source. Ground motion models calibrated on the Joyner-Boore distance are very often used to perform probabilistic seismic hazard analyses, since “mean” depths for seismogenic sources are generally assumed.

Figure 9 shows the predicted acceleration response spectra by the two models for normal (NF, Figs. 9a, 9b, 9e, 9f) and thrust faults (TF, Figs. 9c, 9d, 9g, 9h), two V_{S30} values (e.g. 300 m/s and 600 m/s) and two Joyner-Boore distances, equal to 10km and 30 km, respectively. For the comparison, the prediction of ITA18-ML has been made assuming a hypocentral depth equal to 10 km, which is very close to the event pseudo-depth of ITA18. The spectral shapes allow us to appreciate the difference in the predictions at all frequencies. We selected two reference magnitudes, 4.5 and 6.5, that are very well and scarcely sampled by the dataset, respectively. The larger differences are found for Mw 6.5 for both fault mechanisms (Figs. 9a to 9d), where ITA18-ML predicts larger IMs, in particular in the period range (0.1s – 0.8s). The two GMMs show minor differences, at all periods, for events with a magnitude equal to 4.5 (Figs. 9e to 9h).

Finally, we examine the prediction performance of the two models for three events that occurred after 2016, that are not part of the training dataset. One event occurred on 2019-11-07 at 17:35:21 UTC in Southern Italy with a normal faulting mechanism, a focal depth of 16 km and a magnitude Mw = 4.4 (event id = EMSC-20191107_0000114 in http://esm-db.eu). The event has been recorded by 120 stations and we predict two intensity measures, PGA and SA at T = 1s, using ITA18 and ITA18-ML. Figure 10a shows the difference between the observation and predictions of the two models. The machine learning algorithm and ITA18 predict with similar accuracy PGA ($RMSE_{ML}$ = 0.24 and $RMSE_{ITA18}$ = 0.21, respectively) and SA at T = 1 s ($RMSE_{ML}$ = 0.24 and $RMSE_{ITA18}$= 0.22) with a slightly better performance of ITA18.

The same occurs for the event of 2022-11-09 at 06:07:25 UTC (event id = INT-20221109_0000046 in http://esm-db.eu), a compressive earthquake in the central Adriatic sea with Mw = 5.5. In this case, both predictive models tend to underpredict the observations with a similar performance. Finally, both models tend to overpredict the PGA of a very recent event that occurred on 2023-03-09 at 19:08:05 UTC (event id = INT-20230309_0000266 in http://esm-db.eu), probably because of very fast attenuation of high frequency ground motion with distance.

Discussion and conclusions

In this study, we examine the advantages and disadvantages of the use of a functional form versus machine learning algorithms in ground-motion models. The dataset selected for the analysis is the Lanzano et al (2019), used to derive the most recent ground motion model for Italy (ITA18). The dataset consists of 5,607 records, relative to 146 earthquakes and 1657 stations, which is comparable, in terms of the number of waveforms, to the datasets used recently to calibrate models based on ML techniques worldwide.

We have selected the most usual variables to calibrate GMMs with classical regression algorithms (e.g. magnitude, distance, hypocentre depth, style of faulting, V_{S30}) to better compare the most recent model for Italy (ITA18) to our results (ITA18-ML) and explain the differences. The ML algorithm with the best predicting capability is the Gaussian Process Regression available in Matlab, which significantly reduces the standard deviation of the predictions.

Machine Learning allows for a reduction of the total standard deviation by about 30% for peak values and spectral acceleration and, in particular, the largest reduction (about 50%) is observed for the between-event standard deviation. The site-to-site standard deviation, although reduced, is affected by the largest variability.

We found that the largest difference in ground motion prediction between models when the event hypocentral depth is considered a separate variable. In particular, in the case of deep events (e.g. depth > 10km) the predicted ground motion in the near-fault has lower values than traditional models (e.g. ITA18) and higher values at long distances, probably because the machine learning algorithm is successful in accounting for the effect of multiple reflections of seismic waves.

The size of the dataset is also crucial: our results suggest that machine learning and traditional regression have similar outcomes in the case of large datasets that adequately sample the variable space. Major differences arise when the sampling set is scarcely populated. Traditional regression-based models show a better performance than ML algorithms when the amount of observations is limited since they compensate the lack of data with predefined linear or nonlinear equations that are based on physical considerations. Therefore the use of traditional regression is more appropriate when the available dataset is poor, like in the case of this study, although the uncertainty about the mean (e.g. standard deviation) could be larger than machine learning approaches.

When large datasets will be available worldwide, of the order of several thousands of records (actually only Japanese datasets are sufficiently large), the advantage of ML approaches could be more attractive and the use of variables that are not directly related to physics-based phenomena (such as event depth, coordinates of seismic stations, station backazimuth or others) that could be of great help in increasing the prediction performance.

Declarations

Funding

This work was supported by the Centro di Pericolosità Sismica (CPS) of Istituto Nazionale di Geofisica e Vulcanologia

Competing Interests

Authors Luzi, Lanzano and Felicetta declare they have no financial interests

Author Contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Lucia Luzi and Chiara Felicetta. The first draft of the manuscript was written by Lucia Luzi and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

The dataset analysed during the current study is available in the *ITA18 strong-motion flat-files* repository, https://doi.org/10.13127/ita18/sa_flatfile.

References

1. Atik Al L, Abrahamson NA, Bommer JJ, Scherbaum F, Cotton F, Kuehn N (2010) The variability of ground-motion prediction models and its components. Seismological Research Letters, 1–8. http://doi.org/10.1785/gssrl.81.5.794.
2. Campbell KW, Bozorgnia Y (2014) NGA-West2 Ground Motion Model for the Average Horizontal Components of PGA, PGV, and 5% Damped Linear Acceleration Response Spectra. Earthquake Spectra, 30(3), 1087–1115. http://doi.org/10.1193/062913EQS175M
3. Chiou BSJ, Youngs RR (2014) Update of the Chiou and Youngs NGA Model for the Average Horizontal Component of Peak Ground Motion and Response Spectra. Earthquake Spectra, 30(3), 1117–1153. http://doi.org/10.1193/072813EQS219M
4. Derakhshani A, Foruzan AH (2019) Predicting the principal strong ground motion parameters: A deep learning approach. Appl. Soft Comput. 80, 192–201, doi: 10.1016/j.asoc.2019.03.029
5. Derras, B, Bard PY, Cotton F, Bekkouche A (2012) Adapting the neural network approach to PGA prediction: an example based on the KiK-net data. Bull. Seismol.Soc. Am. 102 (4), 1446–1461.
6. Douglas J, Edwards B (2016) Recent and future developments in earthquake ground motion estimation. Earth Science Reviews, 160(C), 203–219. http://doi.org/10.1016/j.earscirev.2016.07.005
7. Güllü H, Erçelebi E (2007) A neural network approach for attenuation relationships: An application using strong ground motion data from Turkey. Engineering Geology, Volume 93, Issues 3–4, Pages 65-81, https://doi.org/10.1016/j.enggeo.2007.05.004.

8. Kerh T, Ting SB (2005) Neural network estimation of ground peak acceleration at stations along Taiwan high-speed rail system. Eng. Appl. Artif. Intell. 18 (7), 857–866.
9. Kim S, Hwang Y, Seo H, Kim B (2020) Ground motion amplification models for Japan using machine learning techniques. Soil Dynam. Earthq. Eng. 132, 106095, doi: 10.1016/j.soildyn.2020.106095.
10. Khosravikia F, Clayton P (2021) Machine learning in ground motion prediction. Computers & Geosciences. 148, 104700. http://doi.org/10.1016/j.cageo.2021.104700
11. Lanzano G, Luzi L, Pacor F, Felicetta C, Puglia R, Sgobba S, D'Amico M (2019) A Revised Ground-Motion Prediction Model for Shallow Crustal Earthquakes in Italy. Bulletin of the Seismological Society of America, 109(2), 525–540. http://doi.org/10.1785/0120180210
12. Lanzano G, Sgobba S, Luzi L, Puglia R, Pacor F, Felicetta C, et al. (2018) The pan-European Engineering Strong Motion (ESM) flatfile: compilation criteria and data statistics. 1–22. http://doi.org/10.1007/s10518-018-0480-z
13. Lanzano G., Ramadan F., Luzi L., Sgobba S., Felicetta C., Pacor F., D'Amico M., Puglia R., Russo E. (2022). Parametric table of the ITA18 GMM for PGA, PGV and Spectral Acceleration ordinates. Istituto Nazionale di Geofisica e Vulcanologia (INGV). https://doi.org/10.13127/ita18/sa_flatfile
14. Luzi L, Pacor F, Puglia R (2020) Italian Accelerometric Archive v3.0, Istituto Nazionale di Geofisica e Vulcanologia, Dipartimento della Protezione Civile Nazionale, https://doi.org/10.13127/itaca.3.0, 2019.
15. Luzi L, Lanzano G, Felicetta C, D’Amico MC, Russo E, Sgobba S, Pacor F, and ORFEUS Working Group 5 (2020) Engineering Strong Motion Database (ESM), Version 2.0. Istituto Nazionale di Geofisica e Vulcanologia (INGV), https://doi.org/10.13127/ESM.2
16. Mori F, Mendicelli A, Falcone G, Acunzo G, Spacagna RL, Naso G, Moscatelli M (2022) Ground motion prediction maps using seismic-microzonation data and machine learning. Nat. Hazards Earth Syst. Sci., 22, 947–966. http://doi.org/10.5194/nhess-22-947-2022
17. Okazaki T, Morikawa N, Fujiwara H, Ueda N (2021). Monotonic neural network for ground-motion predictions to avoid overfitting to recorded sites. Seismol. Res. Lett. 92, no. 6, 3552–3564, doi: 10.1785/0220210099.
18. Pacor F, Paolucci R, Luzi L, Sabetta F, Spinelli A, Gorini A, et al. (2011). Overview of the Italian strong motion database ITACA 1.0, 9(6). 1723–1739. http://doi.org/10.1007/s10518-011-9327-6
19. Paolucci R, Pacor F, Puglia R, Ameri G, Cauzzi C, Massa M (2011) Record processing in ITACA, the new Italian strong-motion database. In: Earthquake Data in Engineering Seismology, S. Akkar, P. Gulkan, and T. Van Eck (Editors), Geotechnical, Geological and Earthquake Engineering Series, Vol. 14, Springer, Dordrecht, The Netherlands, 99–113.
20. Rasmussen CE, Williams CKI (2006) Gaussian Processes for Machine Learning. MIT Press. Cambridge, Massachusetts
21. Russo E, Felicetta C, D Amico M, Sgobba S, Lanzano G, Mascandola C, Pacor F, Luzi L (2022) Italian Accelerometric Archive v3.2, Istituto Nazionale di Geofisica e Vulcanologia, Dipartimento della Protezione Civile Nazionale. doi: 10.13127/itaca.3.2

22. Seo H, Kim J, Kim B (2022) Machine-Learning-Based Surface Ground- Motion Prediction Models for South Korea with Low-to-Moderate Seismicity. Bulletin of the Seismological Society of America, 112(3), 1549–1564, http://doi.org/10.1785/0120210244

Figures

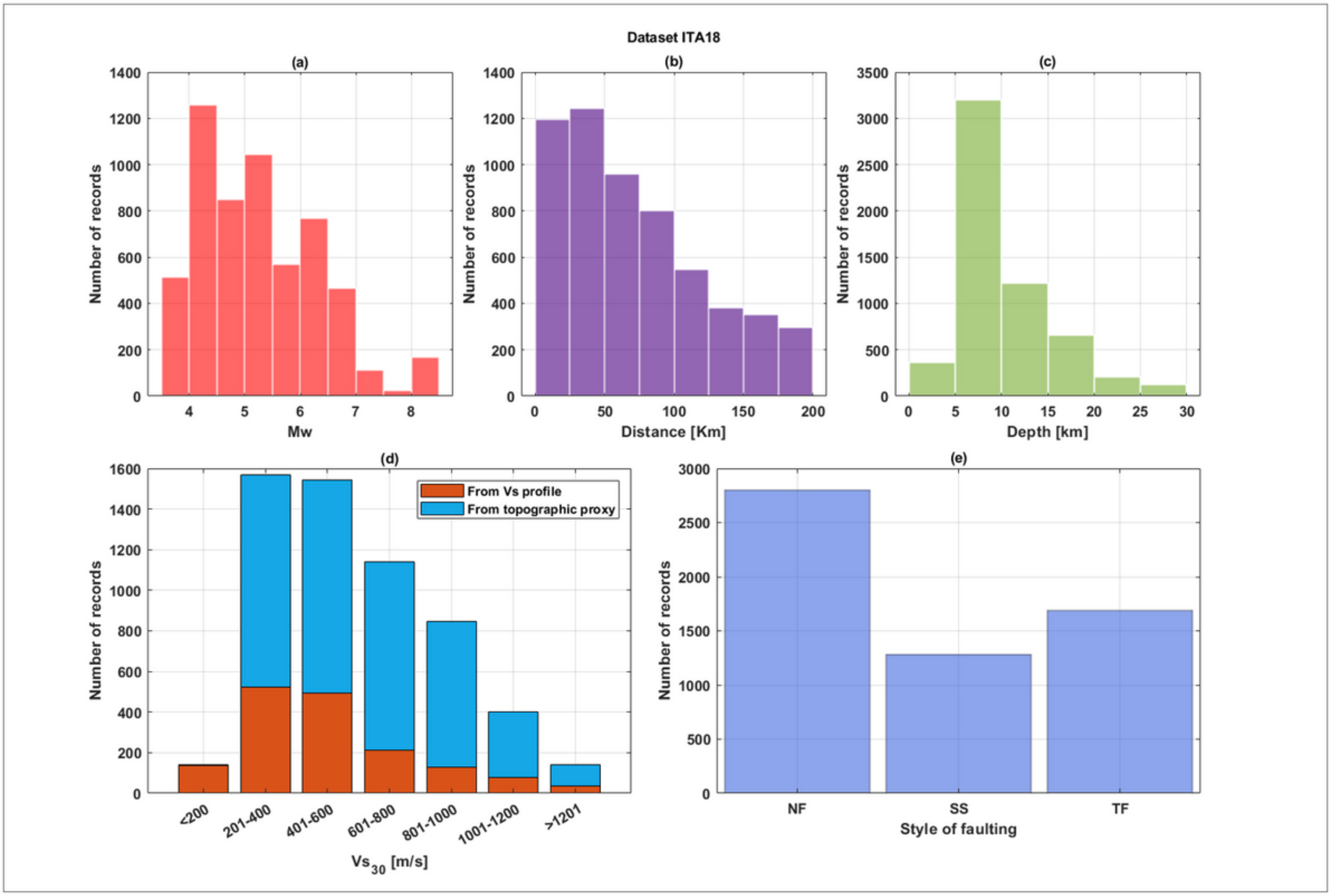


Figure 1

Distribution of the metadata of the seismic records used for the analysis.

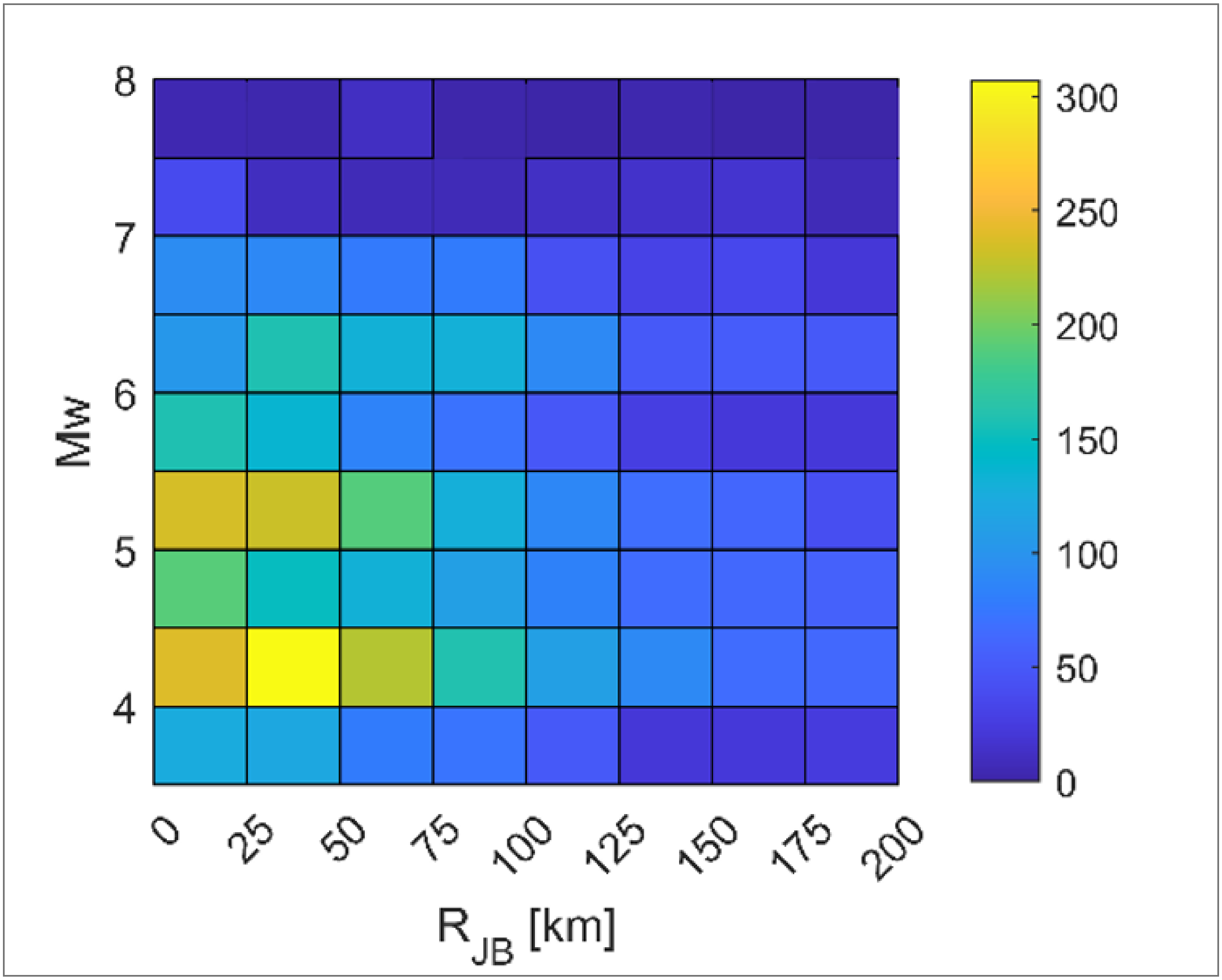
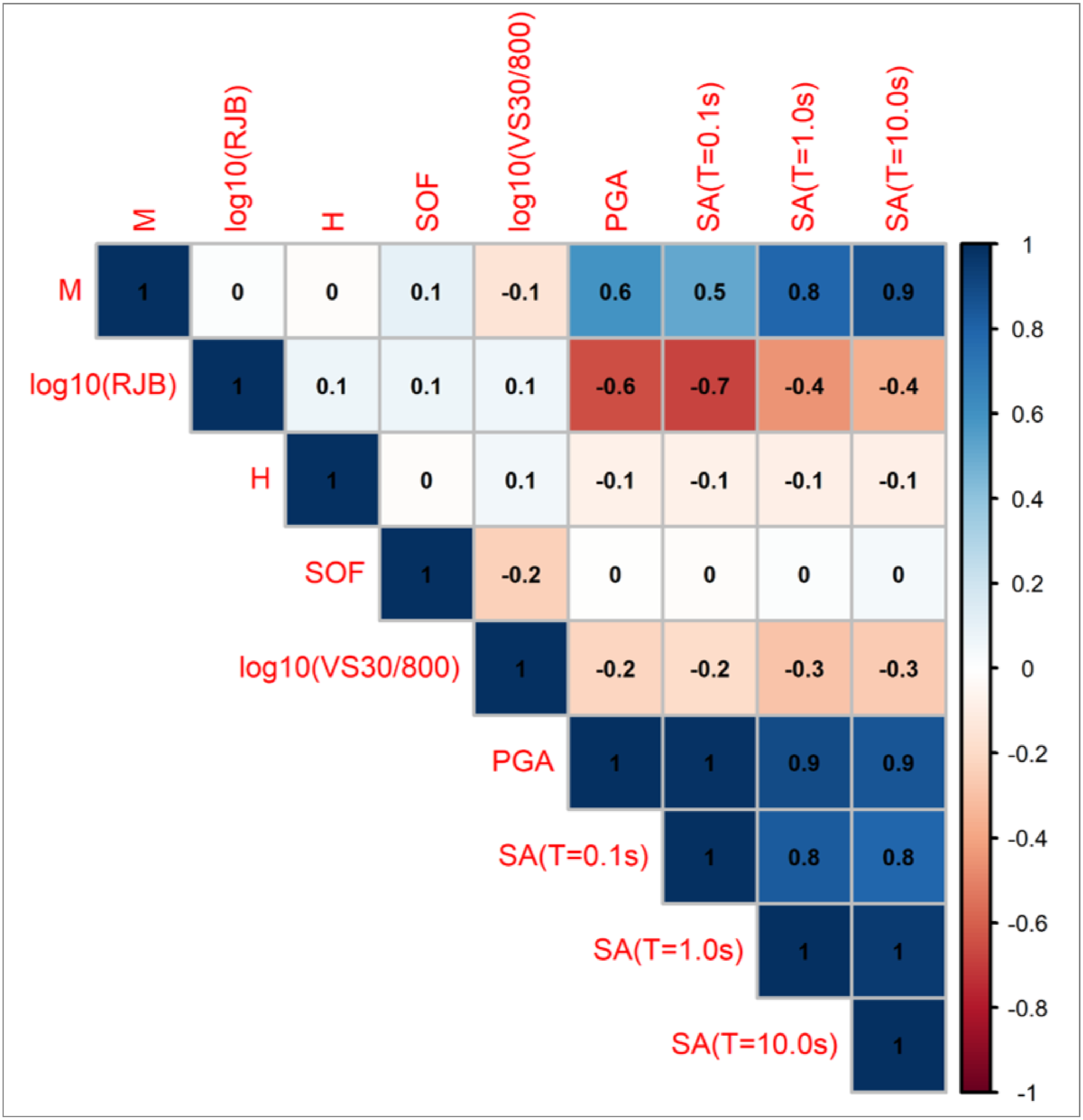


Figure 2

Record density for magnitude–distance combinations.



	M	log10(RJB)	H	SOF	log10(VS30/800)	PGA	SA(T=0.1s)	SA(T=1.0s)	SA(T=10.0s)
M	1	0	0	0.1	-0.1	0.6	0.5	0.8	0.9
log10(RJB)		1	0.1	0.1	0.1	-0.6	-0.7	-0.4	-0.4
H			1	0	0.1	-0.1	-0.1	-0.1	-0.1
SOF				1	-0.2	0	0	0	0
log10(VS30/800)					1	-0.2	-0.2	-0.3	-0.3
PGA						1	1	0.9	0.9
SA(T=0.1s)							1	0.8	0.8
SA(T=1.0s)								1	1
SA(T=10.0s)									1

Figure 3

Correlation matrix among the predictor variables.

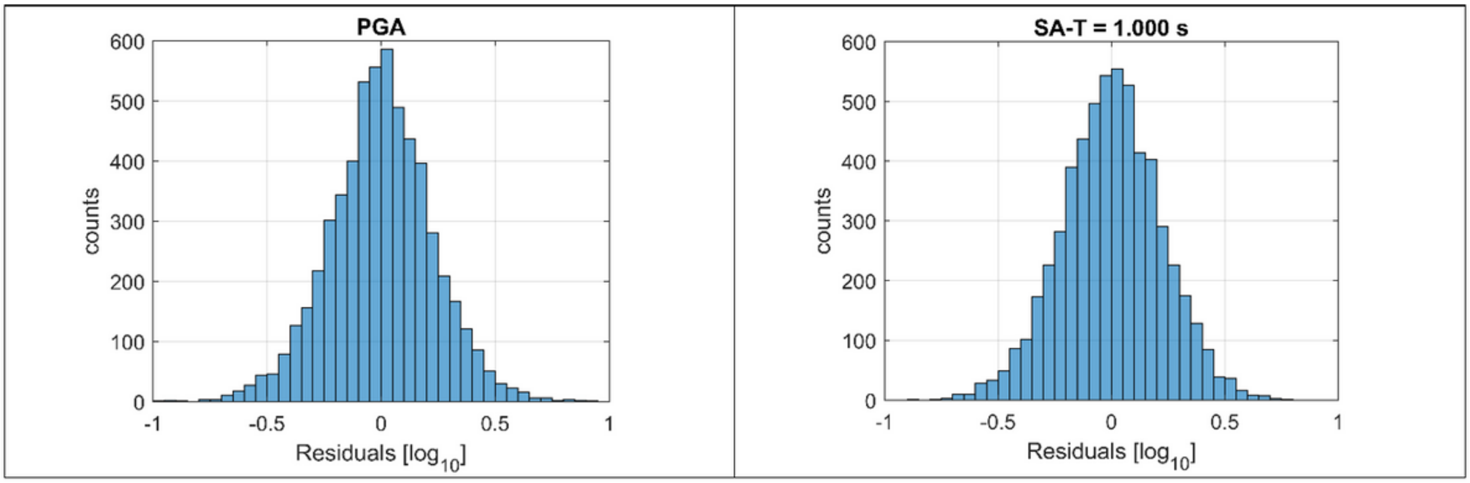


Figure 4

Residual distribution: left Peak Ground Acceleration; right: Spectral Acceleration at T = 1s.

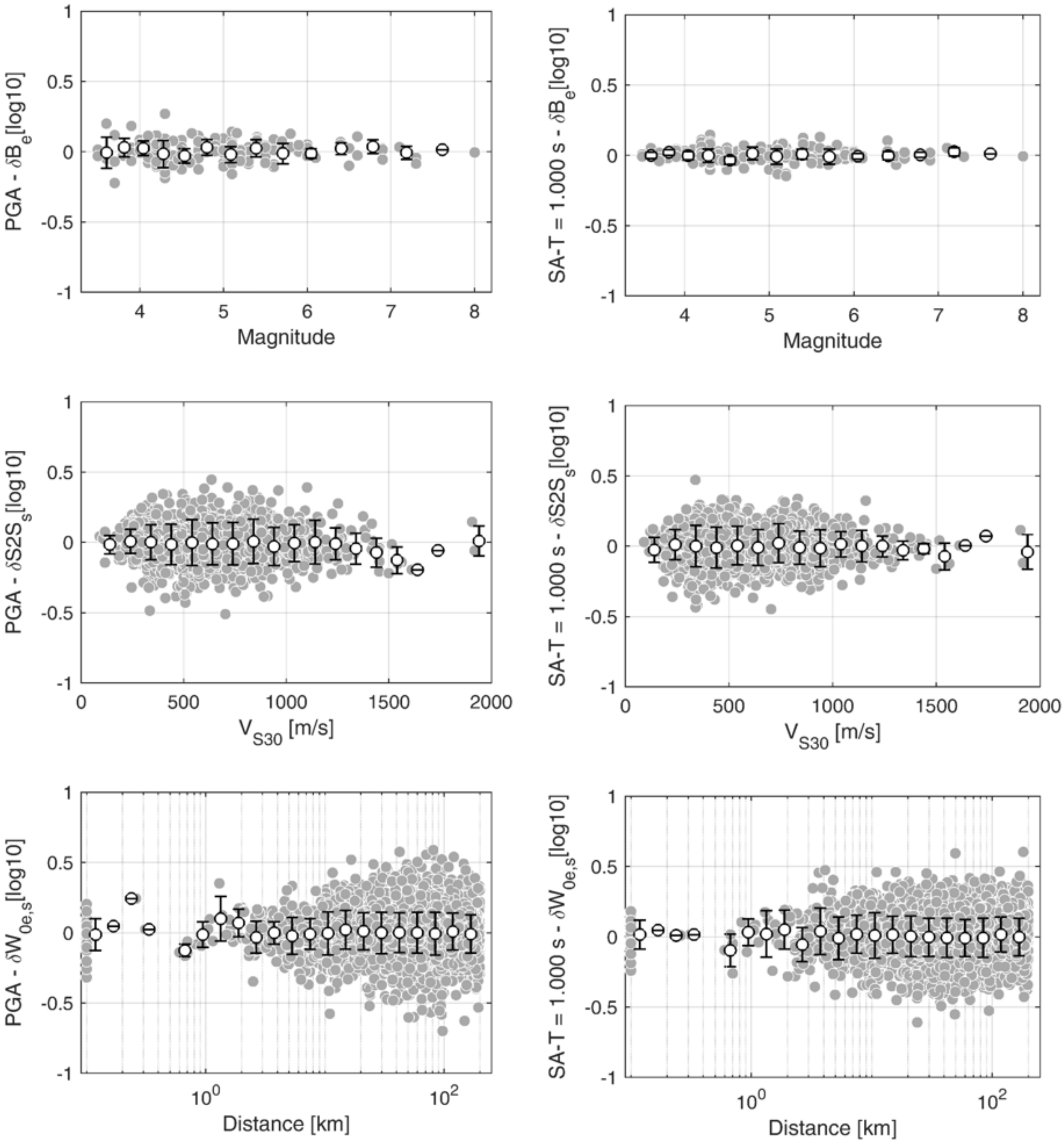


Figure 5

Top panel: Between-event residuals in function of magnitude; middle panel: site-to-site term in function of V_{s30}; bottom panel: event- and site- corrected residuals in function of Joyner-Boore distance. Left column: Peak Ground Acceleration; right column: Spectral Acceleration ordinates at T = 1.0s.

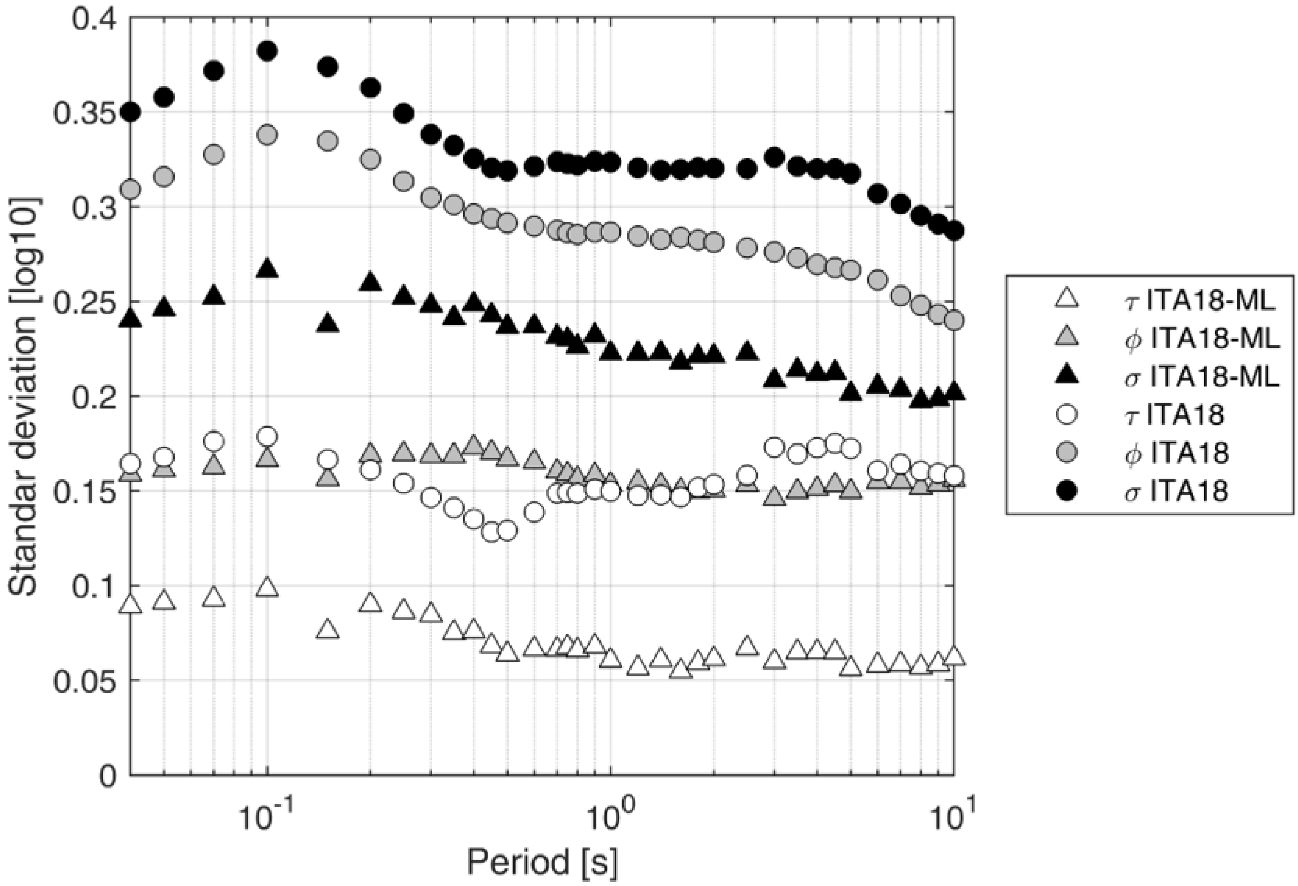


Figure 6

Comparison between ITA18 and ITA18ML in terms of between-event, site-to-site and total standard deviation, in function of the spectral periods.

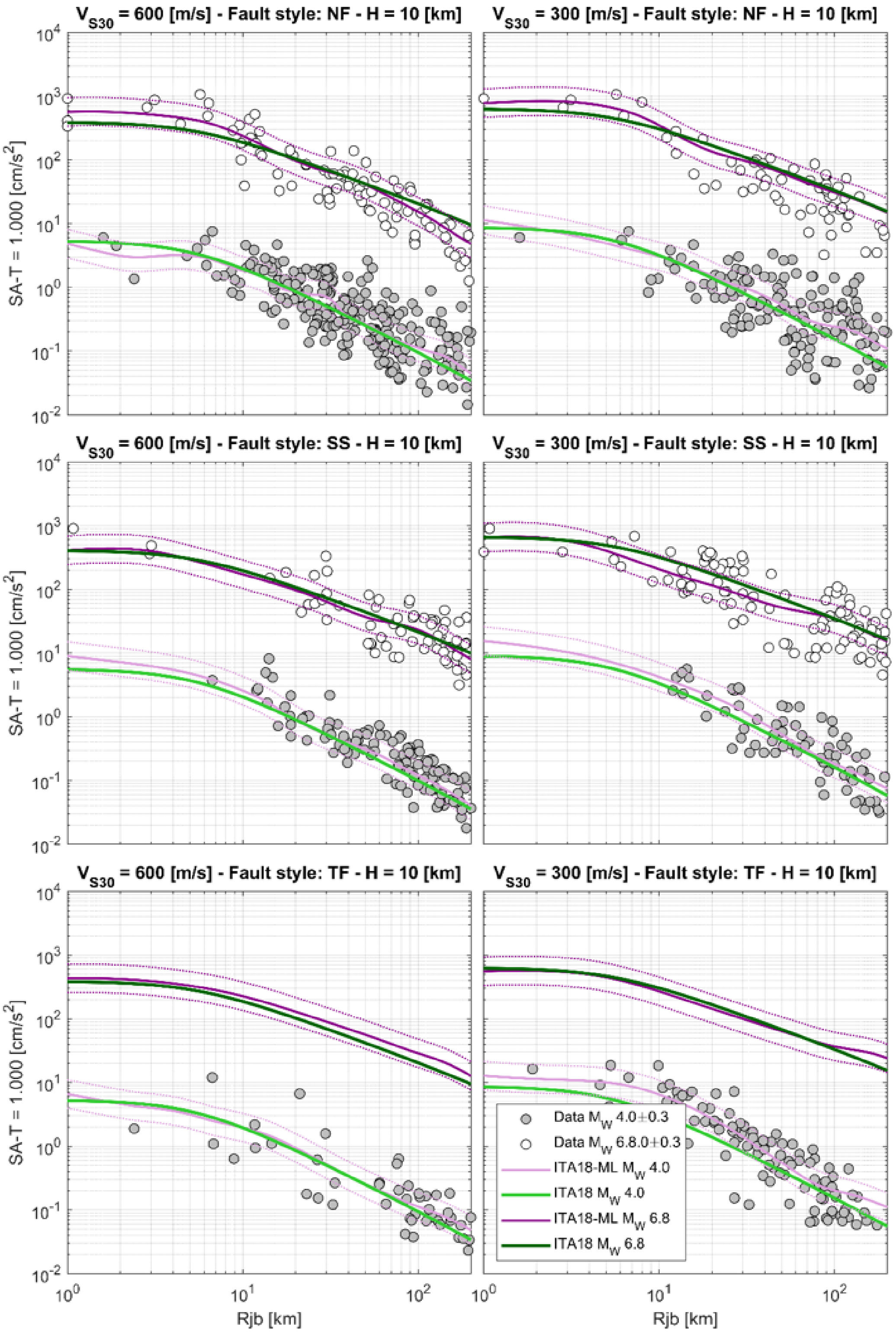


Figure 7

Comparison between the ITA18 and ITA18-ML models for Mw 4.0 and Mw = 6.8 and a hypocentral depth equal to 10 km; white and grey dots are observations. Left column: V_{S30} = 600 m/s; right column: V_{S30} = 300m/s; upper panel: normal fault (NF); middle panel: strike lip faults (SS); bottom panel: thrust faults (TF).

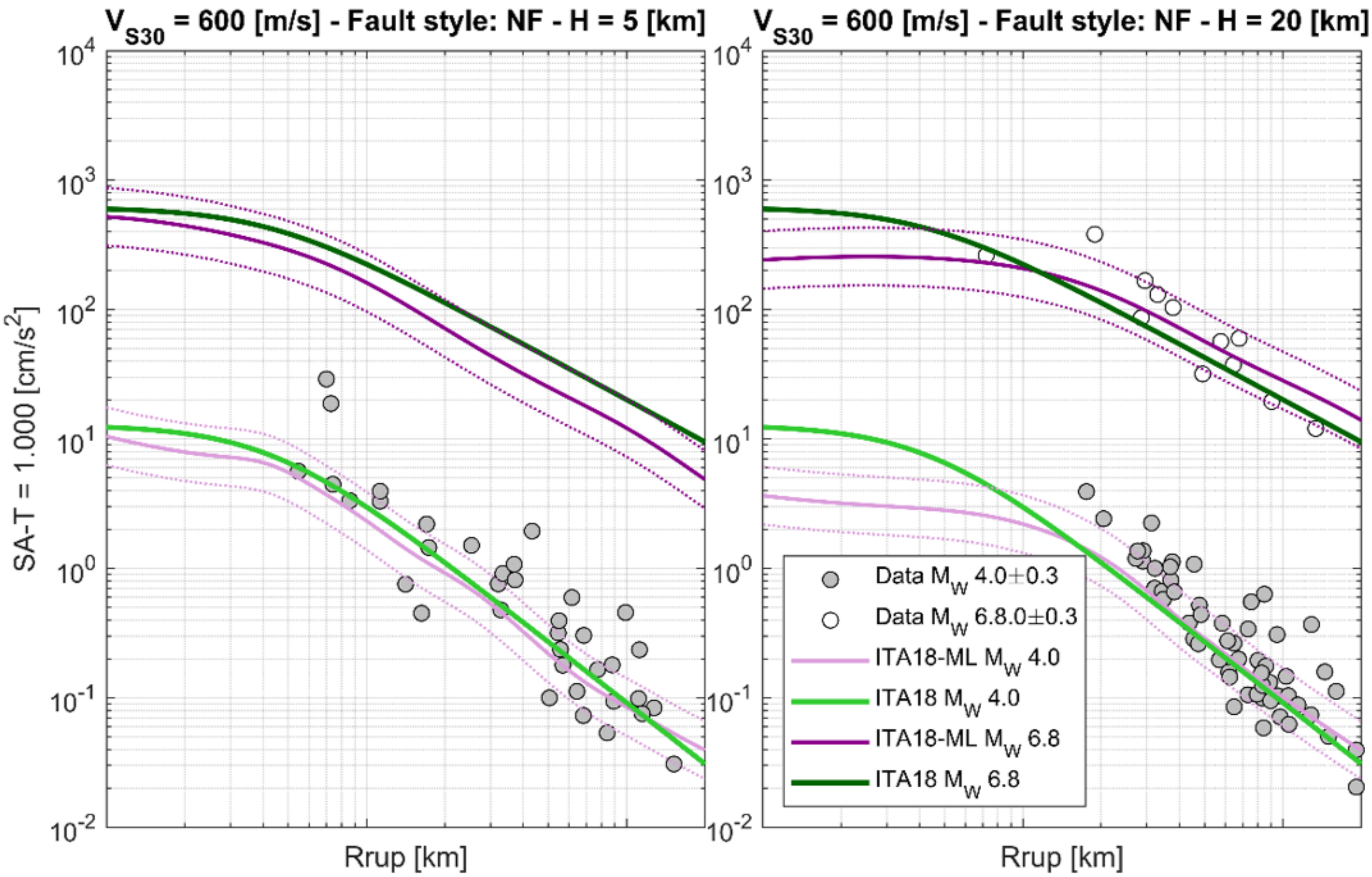


Figure 8

Comparison of ITA18 and ITA-ML for Mw 4.0 and Mw = 6.8, normal style of faulting and V_{S30} = 600 m/s and two hypocentral depths; left: H = 5 km; right: H = 20 km; the ITA18 prediction has been inferred using the rupture distance.

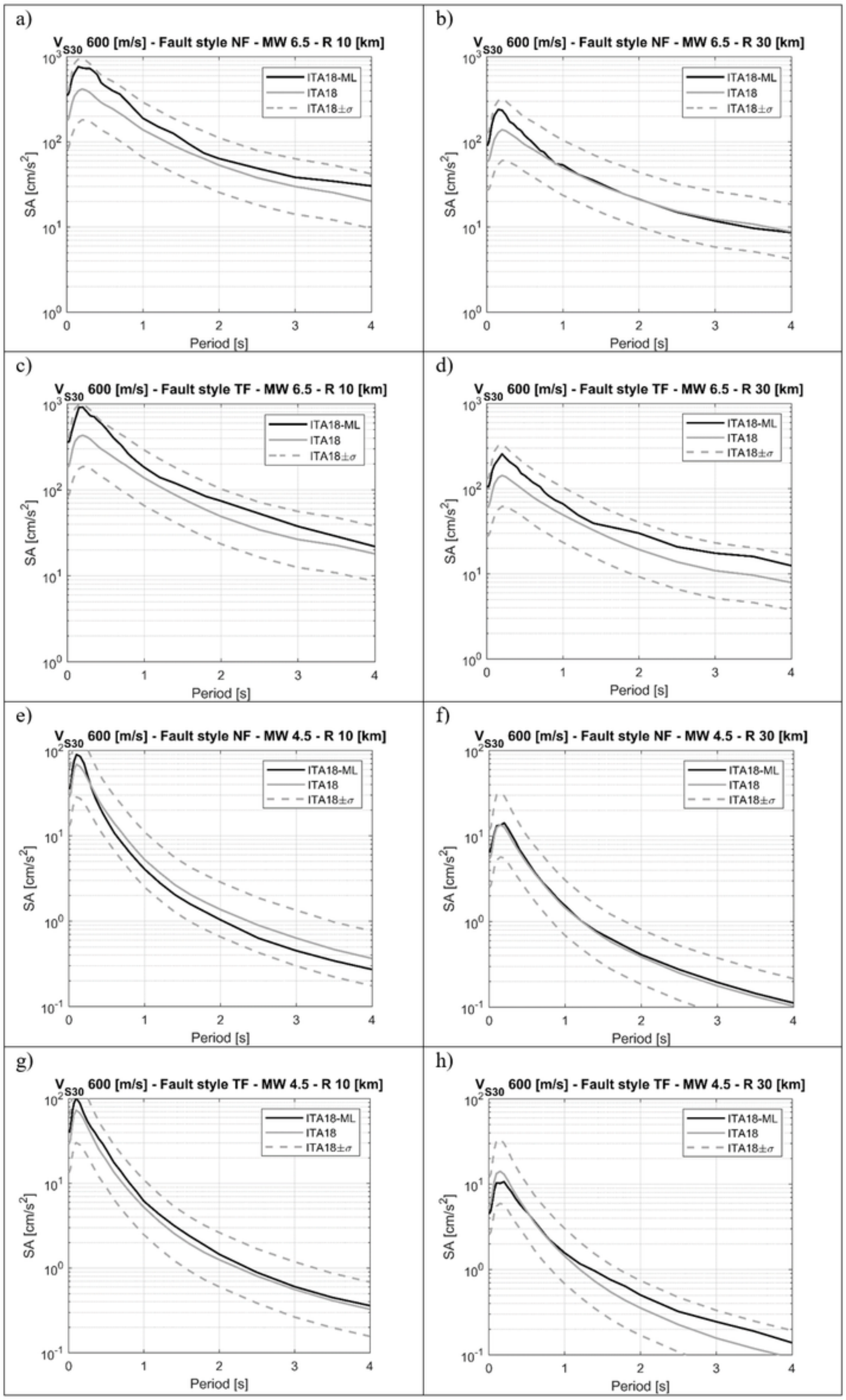


Figure 9

Comparison between the acceleration response spectra predicted by ITA18 and ITA-18-ML; left panel: Joyner Boore distance = 10 km; right panel: Joyner Boore distance equal to 30 km; NF = normal fault; TF = thrust fault.

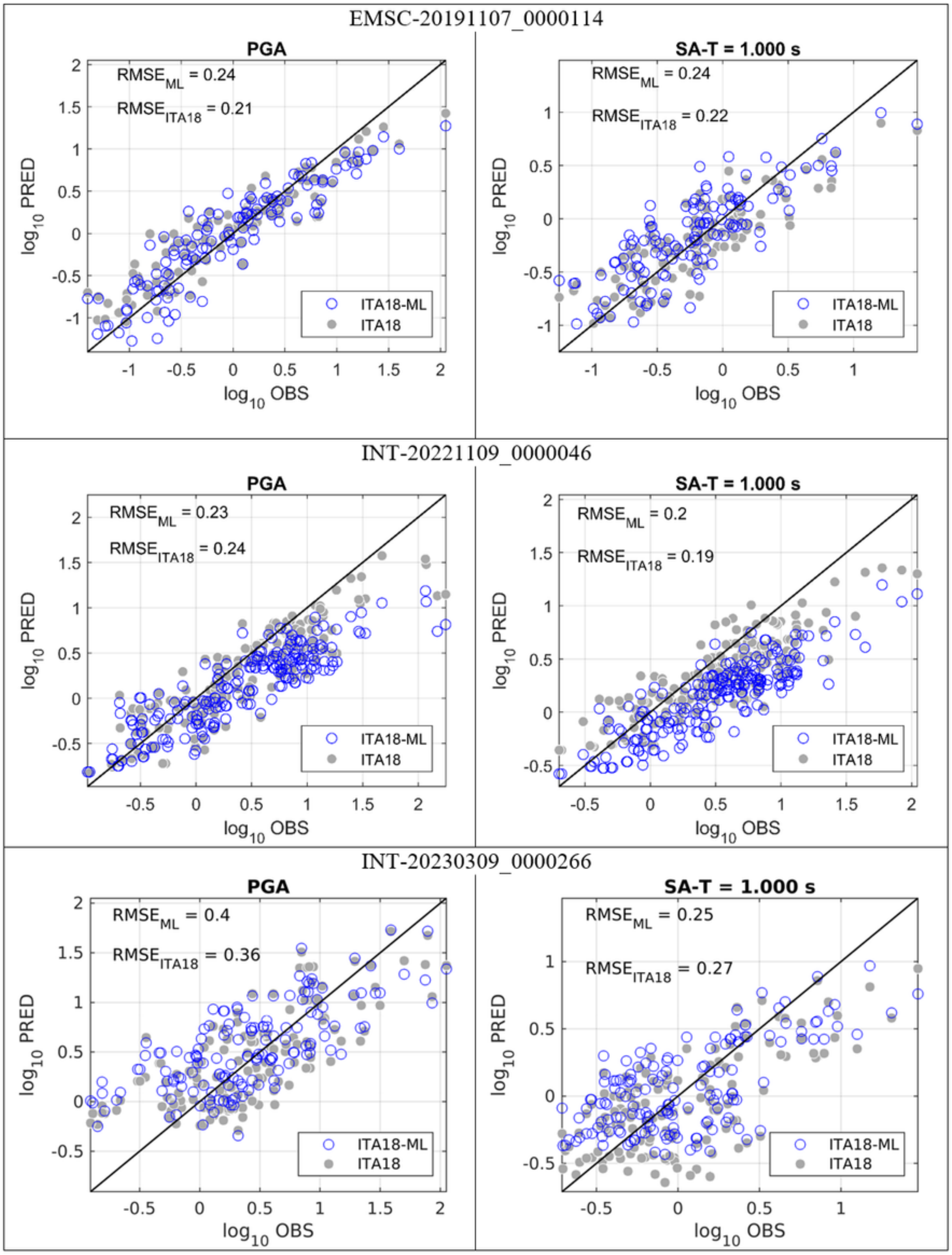


Figure 10

Comparison between observation and predictions for three events EMSC-20191107_0000114, INT-20221109_0000046, and INT-20230309_0000266 in http://esm-db.eu; Left: Peak Ground Acceleration; right: Spectral Acceleration at T = 1s.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- AppendixA.docx